

Tutorial 4 – Create Tile Movement & Shuffling/ Sorting Tiles

First, create a script folder and create a new C# script with the name 'Tile Movement' as it'll manage the movement behaviour for each tile in the jigsaw puzzle. Then, declare a public property called 'Tile' as it represents the specific tile will control its movement.

Depth Sorting of Tiles

Shuffling of Jigsaw Tiles

Final Script:
Tile Movement

```

using UnityEngine;
using UnityEngine.EventSystems;

@ Unity Script | 0 references
public class TileMovement : MonoBehaviour
{
    2 references
    public Tile tile { get; private set; }
    private Vector3 mOffset = new Vector3(0.0f, 0.0f, 0.0f);

    2 references
    private Vector3 GetCorrectPosition()
    {
        return new Vector3(tile.xIndex * mOffset.x, tile.yIndex * 100f, 0f)
    }

    @ Unity Message | 0 references
    private void OnMouseDown()
    {
        if(EventSystem.current.IsPointerOverGameObject())
        {
            return;
        }

        mOffset = transform.position - Camera.main.ScreenToWorldPoint(new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f));
    }

    @ Unity Message | 0 references
    private void OnMouseDrag()
    {
        if (EventSystem.current.IsPointerOverGameObject())
        {
            return;
        }

        Vector3 curScreenPoint = new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f);
        Vector3 curPosition = Camera.main.ScreenToWorldPoint(curScreenPoint) + mOffset;
        transform.position = curPosition;
    }

    @ Unity Message | 0 references
    private void OnMouseUp()
    {
        if(EventSystem.current.IsPointerOverGameObjects())
        {
            return;
        }

        float dist = (transform.position - GetCorrectPosition()).magnitude;
        if(dist < 50f)
        {
            transform.position = GetCorrectPosition();
        }
    }
}

```

Sorting Tiles

```

using NUnit.Framework;
using UnityEngine;

@ Unity Script | 1 reference
public class TileSort : MonoBehaviour
{
    private List<SpriteRenderer> mSortIndices = new List<SpriteRenderer>();

    public TileSort
    {
    }

    0 references
    public void Clear()
    {
        mSortIndices.Clear();
    }

    1 reference
    public void Add(SpriteRenderer renderer)
    {
        mSortIndices.Add(renderer);
        SetRenderOrder(renderer, mSortIndices.Count);
    }

    1 reference
    public void Remove(SpriteRenderer renderer)
    {
        mSortIndices.Remove(renderer);
        for (int i = 0; i < mSortIndices.Count; i++)
        {
            SetRenderOrder(mSortIndices[i], i + 1);
        }
    }

    0 references
    private void BringToTop(SpriteRenderer renderer)
    {
        Remove(renderer);
        Add(renderer);
    }

    2 references
    private void SetRenderOrder(SpriteRenderer renderer, int index)
    {
        renderer.sortingOrder = index;
        Vector3 p = renderer.transform.position;
        p.z = -index / 10.0f;
        renderer.transform.position = p;
    }
}

```

Shuffling Tiles